- Marine biomass production originates with primary productivity, which in turn is affected by the availability of sunlight, carbon dioxide, nutrients such as nitrates and phosphates, and trace elements.
- Marginal seas generally exhibit intermediate levels of primary production, with the highest rates found in coastal upwelling regions and the lowest primary production occurring in open ocean regions.
- Hence, the highest biomass production rates occur in coastal upwelling zones, the lowest in open oceans regions, and intermediate rates in marginal seas.

18. Statement 1: CORRECT—Heating by solar energy causes the water to expand. That is why, near the equator the ocean water is about 8 cm higher in level than in the middle latitudes.
    Statement 2: CORRECT—Winds are responsible for both magnitude and direction [Coriolis force also affects direction] of the ocean currents. Example: Monsoon winds are responsible for the seasonal reversal of ocean currents in the Indian ocean.
19. Three factors aid the formation of Counter- Equatorial current
    1. Piling up of water in the western pacific due to trade winds.
    2. The presence of doldrums (equatorial low pressure belt) in between the north equatorial current and the south equatorial current. Doldrums are narrow regions with calm (lower) atmospheric conditions. Such conditions aid the backward movement of piled up western pacific waters.
    3. Piling of water in the western part of oceans due to rotation of earth.
20. Statement 1: CORRECT—Summer Circulation - North Equatorial Current Counter-Equatorial Current are Absent
    In summer, due to the effects of the strong southwest monsoon and the absence of the north-east trades, a strong current flows from west to east, which completely obliterates *the* north equatorial current. Hence, there is *no* counter-equatorial current *as* well.
    Statement 2: CORRECT—The south equatorial current, partly led by the corresponding current of the Pacific Ocean, flows from east to west.